# SARTHAK

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## PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in full-stack and secure production systems (Spring Boot, Flutter, JWT, AES-256, RSA-2048) for healthcare and government platforms. Applying AI/LLM engineering skills, including RAG, embeddings, fine-tuning, vector DBs, chat clients, chat models, and prompt templates, to build intelligent, context-aware, scalable products.

## EXPERIENCE

**Associate Software Engineer, CREDEX Technology** *Jul 2026 – Present*

- Built the eShram worker registration app's dual-layer encryption: AES-256-CBC with PBKDF2 key derivation (65,536 iterations) and RSA-2048/PKCS1 for sensitive payloads.
- Delivered an image-based CAPTCHA system for eShram with server-side AES-encrypted delivery and app-wide screenshot prevention.

**Software Engineer Intern, CREDEX Technology** *Nov 2025 – Jun 2026*

*Paid Internship | Pre-Placement Offer Extended — promoted to Associate Software Engineer after 8 months*

- Worked on production tickets for a healthcare & government platform serving real end-users.
- Implemented double encoding of JWT tokens and platform-level Redis caching to boost security and reduce repeated DB reads.
- Designed and deployed Spring Boot microservices (Item Master, Form Manager) integrated via API Gateway.
- Contributed to mobile security hardening with AES-256-CBC encryption, RSA-2048 key exchange, and UIDAI-compliant biometric auth.

## PROJECTS

**eShram – Government Platform Worker Registration App** *Flutter | CREDEX Technology*

- Implemented dual-layer encryption: AES-256-CBC with PBKDF2 key derivation (65,536 iterations) and RSA-2048/PKCS1 for sensitive payloads.
- Built an image-based CAPTCHA system with server-side AES-encrypted delivery and app-wide screenshot prevention.

**AI-Powered Chat Assistant (RAG)** *Python | LLM APIs*

- Built a Retrieval-Augmented Generation (RAG) pipeline using embeddings and a vector database to power semantic search over documents.
- Designed reusable prompt templates and integrated a chat client with an LLM chat model to generate context-aware, grounded responses.
- Explored fine-tuning workflows to adapt model responses for domain-specific queries, improving answer relevance and consistency.

**Lincare 2.0 – Healthcare Platform Modernization** *Java 17 & Spring Boot*

- Migrated the legacy application from Java 11 to Java 17 and upgraded to Spring Boot 2.7.
- Implemented JWT + SAML auth, platform-level double encoding, Redis caching, and a global error handler.

**AI-Powered Interview Preparation App** *Chat Model | MongoDB*

- Built a platform that generates personalized technical and HR interview questions using a chat model, with prompt templates tuned per role and experience level.
- Implemented answer feedback and progress-tracking analytics, backed by authentication and a MongoDB backend.

## TECHNICAL SKILLS

**Languages:** Java, JavaScript, Dart, Python
**AI / LLM:** RAG, Embeddings, Fine-Tuning, Vector Databases, Chat Models, Chat Clients, Prompt Templates
**Frontend:** Flutter, HTML, CSS, Tailwind, Bootstrap
**Backend:** Spring Boot, Express.js, REST APIs, Microservices
**Databases:** MongoDB, MySQL, Redis
**Security:** AES-256-CBC, RSA-2048, JWT, SAML, PBKDF2
**Tools:** Git, GitHub, Firebase, API Gateway, VS Code, IntelliJ, Render

## EDUCATION

**Raj Kumar Goel Institute of Technology (AKTU)** *2022 – 2026*

*B.Tech in Computer Science (AI & ML) — CGPA: 7.8/10*

**Eklavya Public School (CBSE)** *2022*

*Intermediate (Science) — 80.8%*